ZORIAN THORNTON

Ph.D Candidate, Genome Sciences, Seattle,WA
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EDUCATION

University of Washington - Seattle,WA *Sep. 2019 - Present*
Ph.D in Genome Sciences (Expected Winter 2025/2026)

Virginia Tech - Blacksburg,VA *Aug. 2015 - May 2019*
B.Sc Statistics

Virginia Tech - Blacksburg,VA *Aug. 2015 - May 2019*
B.Sc Computational Modeling and Data Analytics
Minors in Mathematics and Computer Science

TECHNICAL SKILLS

Methodological Expertise

Machine Learning	Supervised/unsupervised methods, deep learning, Bayesian inference, probabilistic programming, anomaly detection, model benchmarking
Statistical Methods	MCMC, variational inference, hierarchical modeling, regularization and dimensionality reduction techniques
Computational Biology	Phylogenetic inference, sequence analysis, antigenic evolution modeling, epidemiological simulation
High-Performance Computing	Parallel computing, distributed workflows, performance optimization

Tools & Frameworks

Languages	Python, R, Java, SQL, MATLAB, bash
ML/DL	PyTorch, TensorFlow, JAX, NumPyro, scikit-learn
Bioinformatics	Augur, Auspice, Biopython, evofr
Data Science Stack	pandas, numpy, scipy, matplotlib, seaborn
Infrastructure	SLURM, CUDA, openMP, Unix/Linux, git, Claude code

RESEARCH AND WORK EXPERIENCE

Fred Hutchison Cancer Research Center *Seattle,WA*
Doctoral Candidate, advised by Frederick Matsen IV *July 2021 - Present*

- Developing end-to-end machine learning pipelines to predict viral fitness and evolution using large-scale biological data
- Designed and implemented biophysically-inspired neural networks (PyTorch) that enable accurate predictions of protein variant phenotypes from experimental data
- Applied statistical methods to analyze complex biological datasets and extract meaningful patterns
- Collaborated with research groups to validate models and improve computational approaches

Adaptive Biotechnologies *Seattle, WA*
Machine Learning Computational Biology Intern *June 2022 - Sept. 2022*

- Implemented and benchmarked semi-supervised learning methods for classification of T-cell receptor specificity using high-throughput sequencing data
- Designed data preprocessing pipelines to handle complex biological sequence data at scale
- Conducted systematic downsampling experiments to optimize model performance across varying data conditions
- Developed visualization tools to communicate technical findings to cross-functional teams

Virginia Tech Department of Statistics *Blacksburg, VA*
Undergraduate Research with Leah Johnson *Aug. 2017 - May 2019*

- Developed predictive environmental models integrating temperature-dependent biological parameters with disease transmission dynamics to forecast Bluetongue virus spread across different climate scenarios
- Built Bayesian hierarchical models using MCMC methods to quantify uncertainty in environmental-disease relationships, incorporating temperature-sensitive vector life history traits and host-pathogen interactions

Nielsen *Chicago, IL*
Data Science Intern *June 2018 - Aug. 2018*

- Implemented statistical framework to identify possible errors in scanned receipt data and implemented pipeline to attempt to correct errors

University of Washington Department of Genome Sciences *Seattle, WA*
Undergraduate Research with Jim Bruce *June. 2017 - Aug. 2017*

- Visualized inter-surface regions of cross-linked proteins and built probabilistic models to quantify competition between proteins for linkage with other proteins to help scientists characterize protein function, discover mutations, and discover protein-protein interactions to tackle molecular challenges such as cancer

PUBLICATIONS

- **Thornton, Z.**, Tran, T., Figgins, M., Huddleston, J., Bedford, T., Matsen IV, F.A., Haddox, H. (2026). Simulating the genetic and antigenic evolution of viruses under selection pressure from host immunity. *Manuscript submitted to Virus Evolution.*
- Yu, T.*, **Thornton, Z.***, Hannon, W., DeWitt, W.S., Radford, C., Matsen IV, F.A. and Bloom, J.D. (2022). A biophyscial model of viral escape from polyclonal antibodies. *Virus Evolution*, 8(2), veac110.
- El Moustaid, F., **Thornton, Z.**, Slamani, H., Ryan, S. and Johnson, LR (2021). Predicting temperature-dependent transmission suitability of bluetongue virus in livestock. *Parasites Vectors*, 14(1), pp.1-14..
- Keller, A., Chavez, J.D., Eng, J.K., **Thornton, Z.** and Bruce, J.E. (2018). Tools for 3D Interactome Visualization. *Journal of proteome research*, 18(2), pp.753-758.

*:These authors contributed equally to this work.

RESEARCH PRESENTATIONS

- Zorian Thornton, *Predicting Disease Progression of Colorectal Cancer via Self-Organizing Maps*, RECOMB 2019, George Washington University, Washington DC, May 2019
- Zorian Thornton, *Modeling Bluetongue Virus via Markov Chain Monte Carlo Methods*, Student Experiential Learning Conference, Virginia Tech, Blacksburg, VA, Apr. 2018

- Zorian Thornton, *Viewing Molecular Interaction Interfaces Through Directed Computational Methods*, Department of Genome Sciences Research Symposium, University of Washington, Seattle, WA, Aug. 2017